NUS-ISS

Problem Solving Using Pattern Recognition

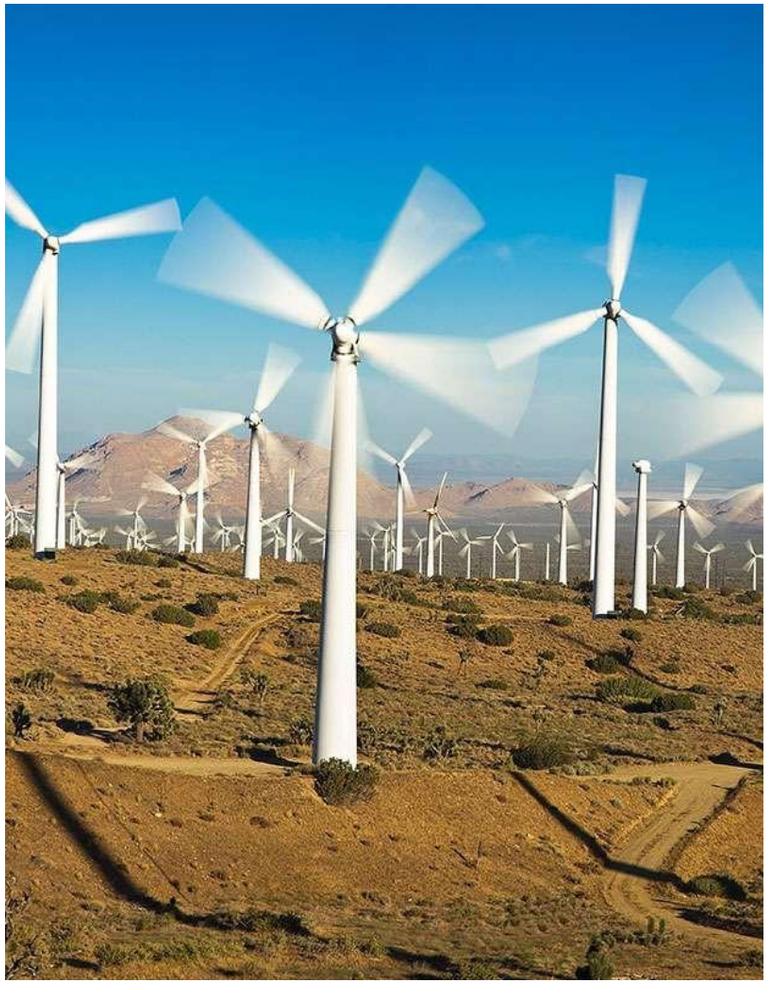
Deep learning: Act

by Zhengqing Hu

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Wind power output prediction

Better economic value



Source: https://deepmind.com/blog/machine-learning-can-boost-value-wind-energy/

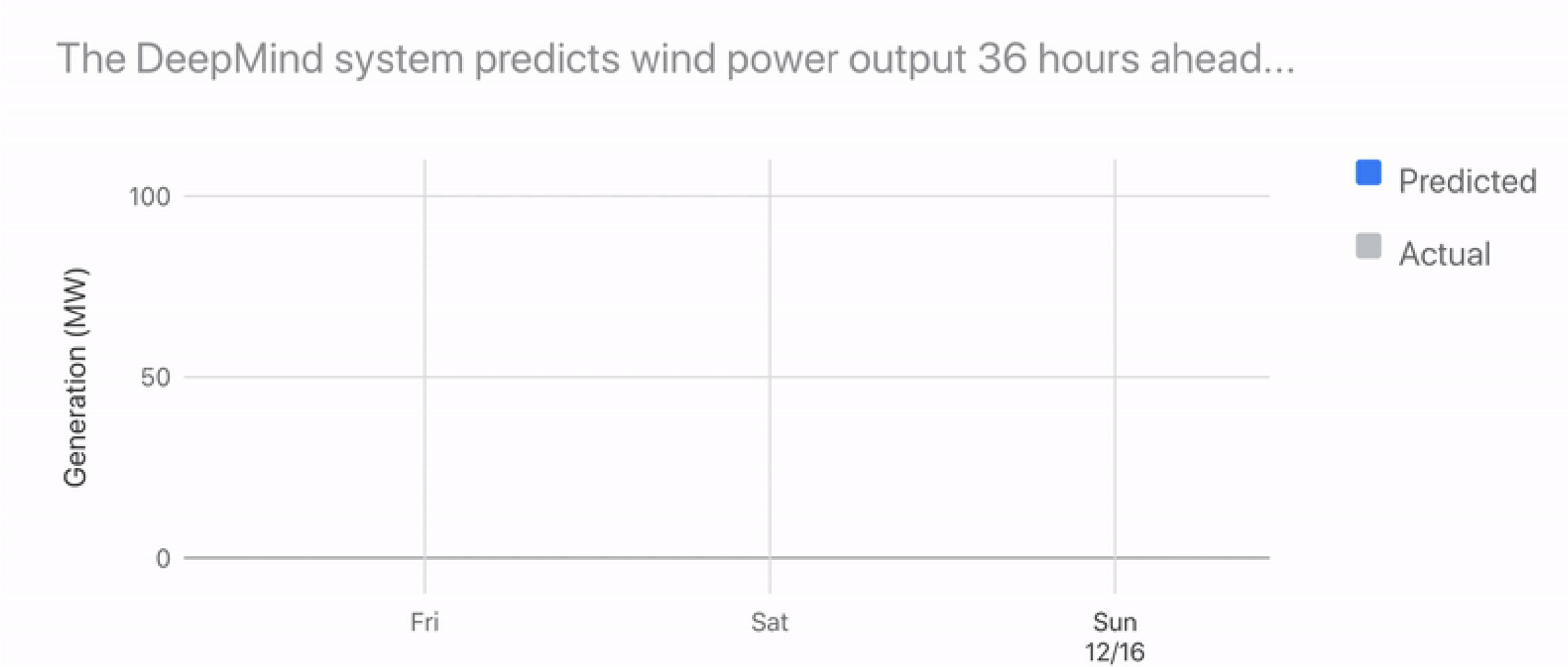
- Use deep learning to predict wind power output ***36*** hours ahead of actual generation
- Based on predictions recommend optimal hourly delivery commitments to power grid a full day in advance
- Benefit: scheduled energy sources are more valuable to grid

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Wind power output prediction

Better economic value

- Actual vs Predicted



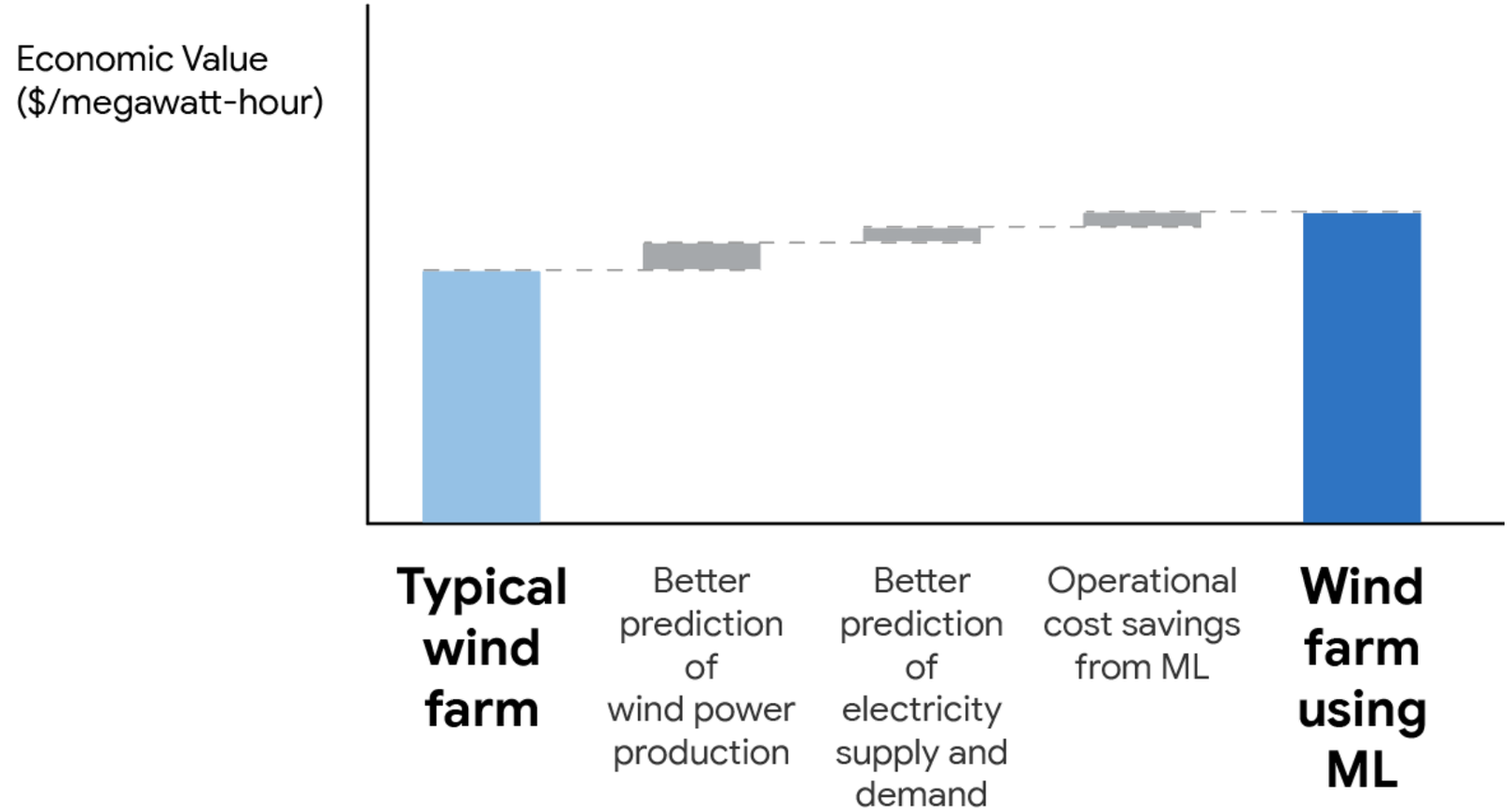
Source: https://deepmind.com/blog/machine-learning-can-boost-value-wind-energy/

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Wind power output prediction

Better economic value

Machine learning can increase the value of wind energy



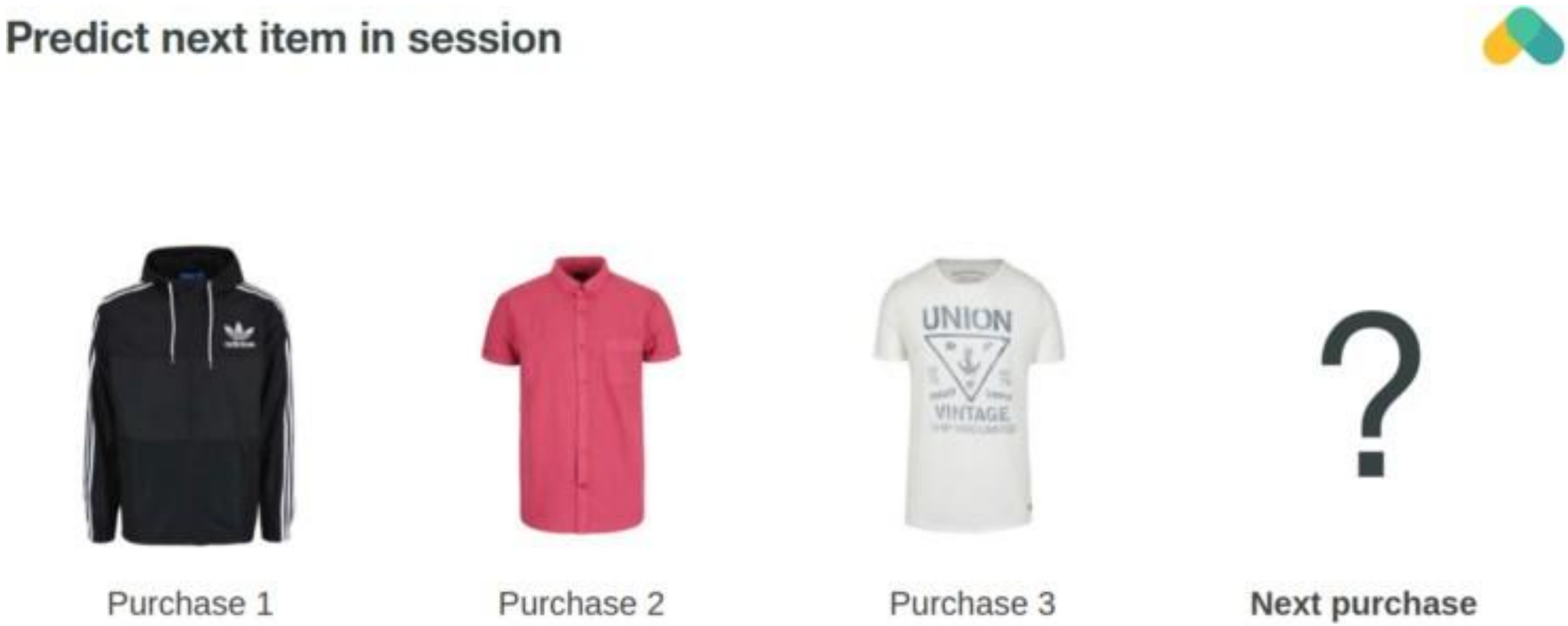
Illustrative results from 2018 Google/DeepMind field study

Source: https://deepmind.com/blog/machine-learning-can-boost-value-wind-energy/

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Predict then recommend

What's next?



Source: https://medium.com/recombee-blog/machine-learning-for-recommender-systems-part-2-deep-recommendation-sequence-prediction-automl-f134bc79d66b

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Recommend based on distance

This is how CNN sees the relations based on identification



Source: https://medium.com/recombee-blog/machine-learning-for-recommender-systems-part-2-deep-recommendation-sequence-prediction-automl-f134bc79d66b

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Recommend based on distance

Poster recommendation based on identification



Source: https://medium.com/recombee-blog/machine-learning-for-recommender-systems-part-2-deep-recommendation-sequence-prediction-automl-f134bc79d66b

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Freekick please

Fifa 18

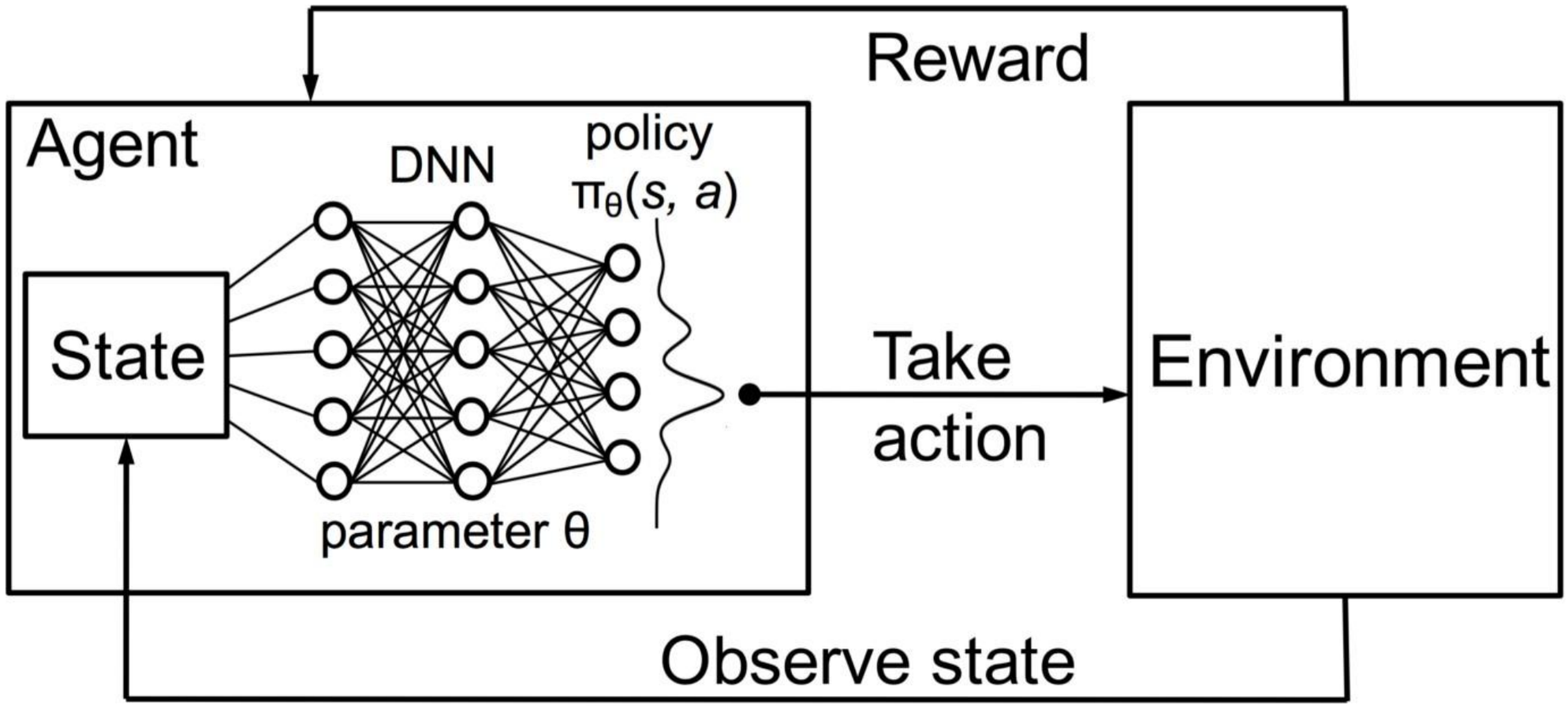


Source: https://towardsdatascience.com/using-deep-q-learning-in-fifa-18-to-perfect-the-art-of-free-kicks-f2e4e979ee66

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Deep Reinforcement learning

Deep Q-learning

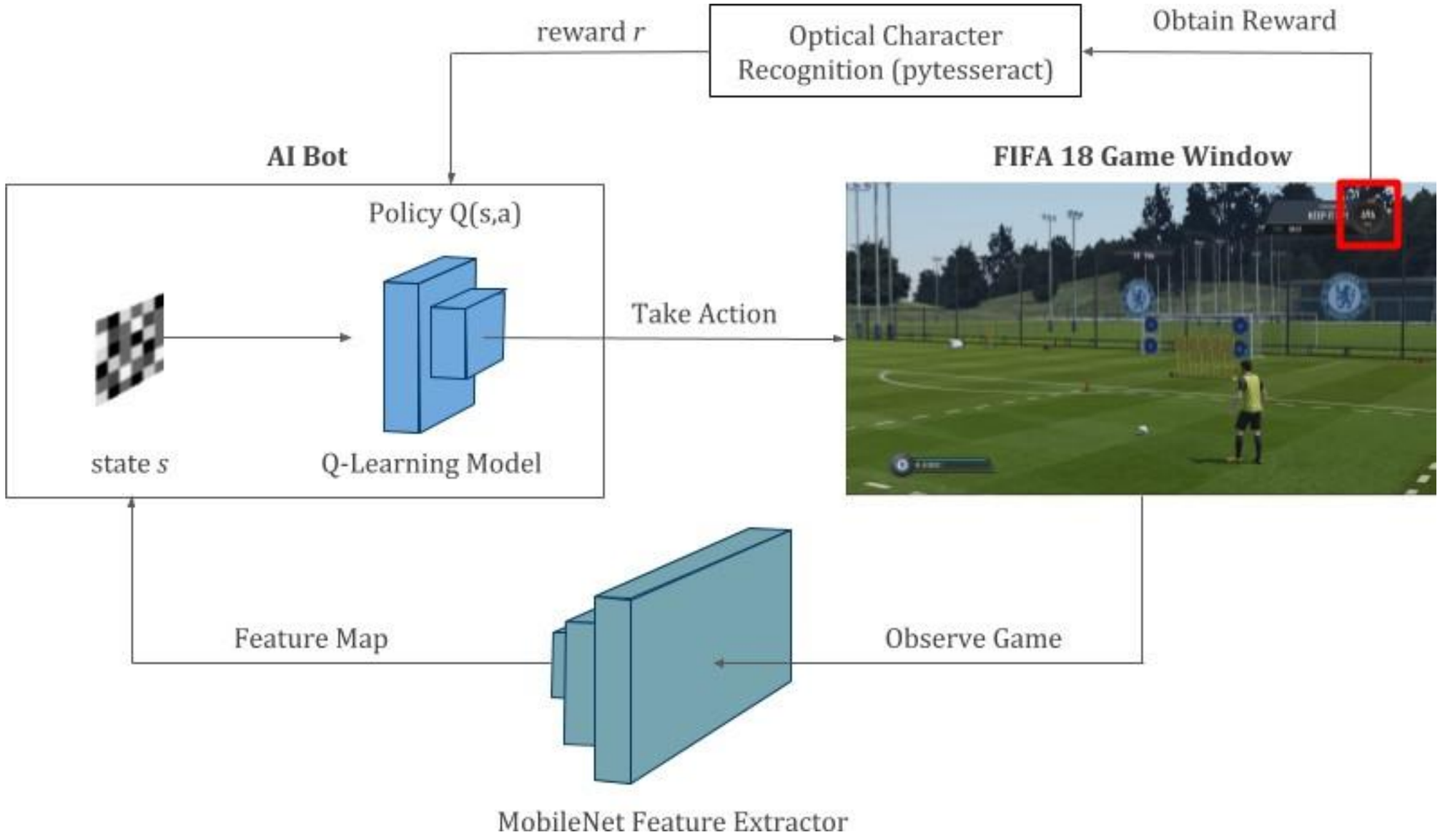


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Deep Reinforcement learning

Fifa 18

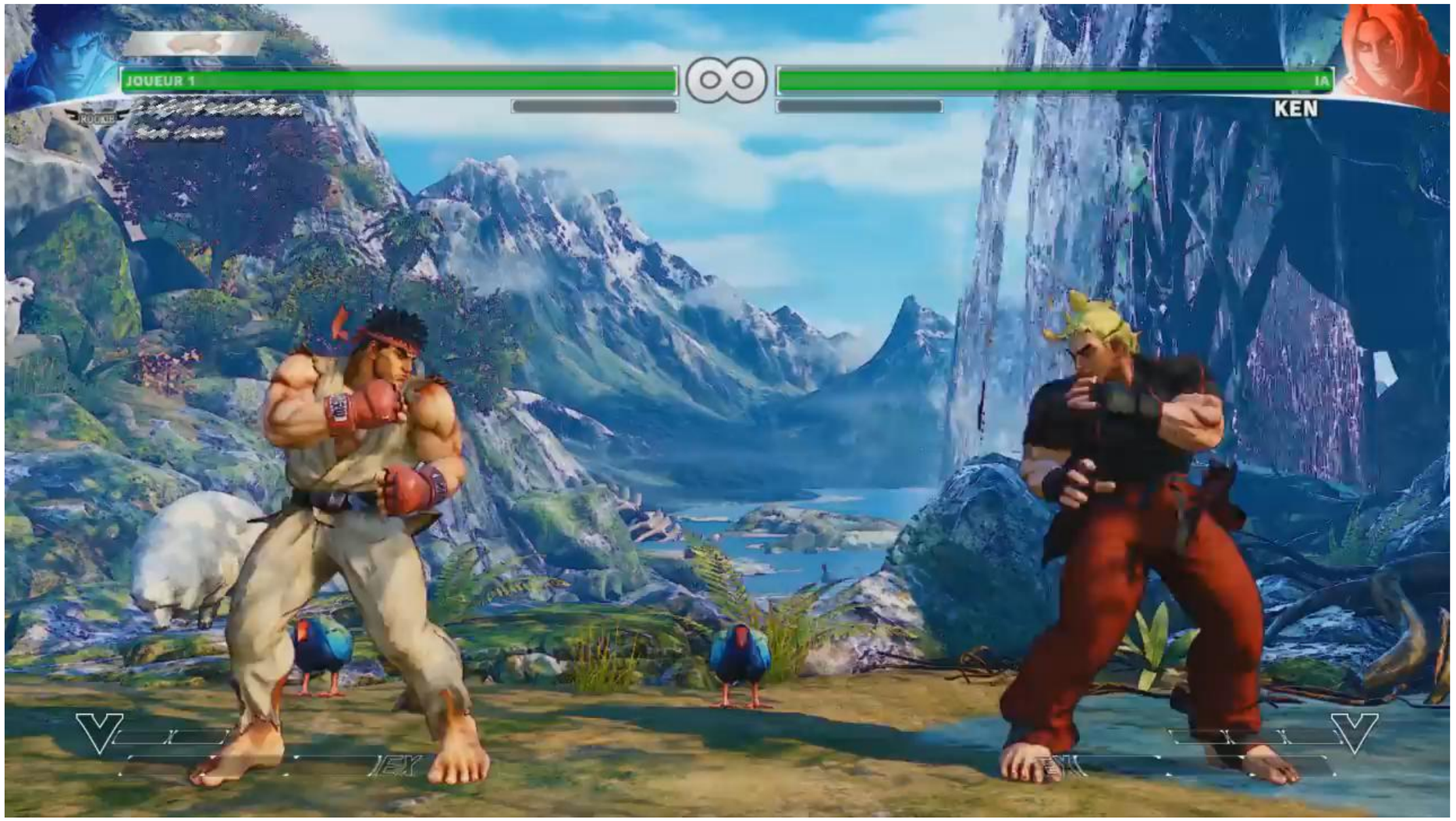


Source: https://towardsdatascience.com/using-deep-q-learning-in-fifa-18-to-perfect-the-art-of-free-kicks-f2e4e979ee66

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AI vs AI

Ryu is played by AI that learned by itself to play



Source: https://www.youtube.com/watch?v=Ao1yJszPQAM

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Throwing

Robot follows

1.29 onwards



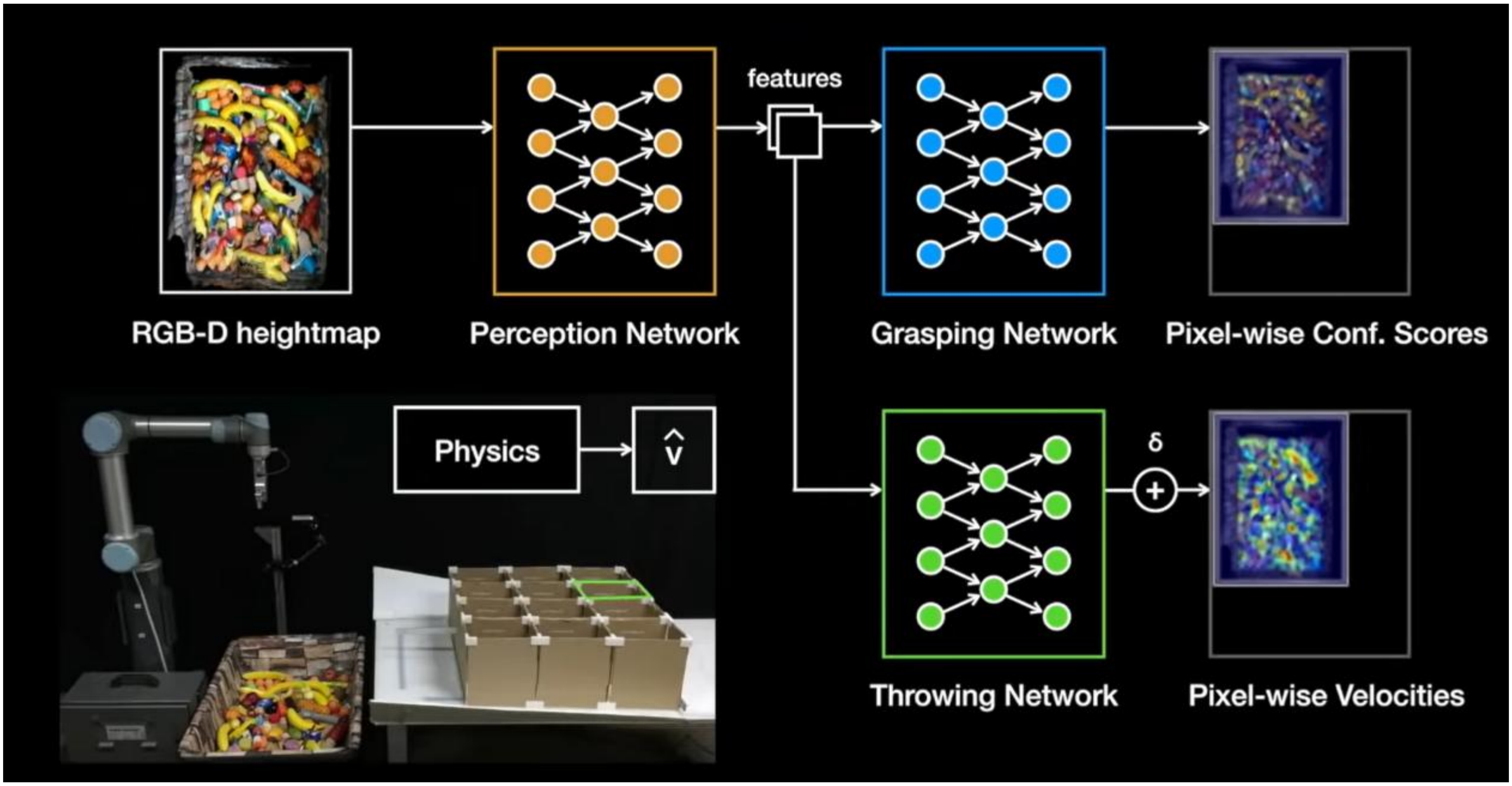
Source: https://x.com/andyzeng_/status/1110655184642936832?mx=2; https://youtu.be/-O-E1nFm6-A

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Throwing

Robot follows

1.29 onwards



Source: https://x.com/andyzeng_/status/1110655184642936832?mx=2; https://youtu.be/-O-E1nFm6-A

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Grasping

Shared learning: Sharing Experience across robots can accelerate learning!

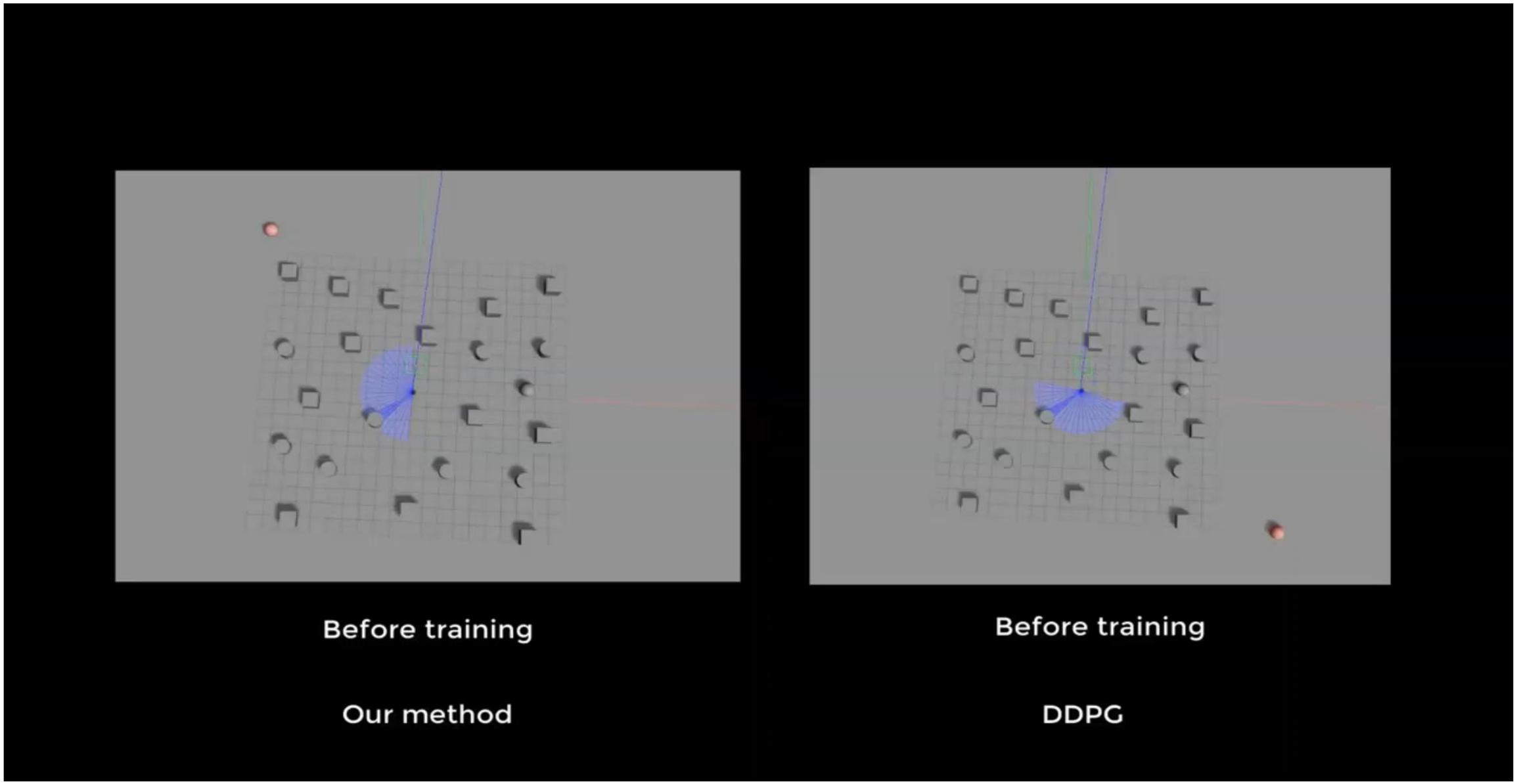


Source: https://ai.googleblog.com/2018/06/scalable-deep-reinforcement-learning.html

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Mobile Robot

Learning how to navigate while avoiding collisions by itself
[Deep Reinforcement Learning (DRL) vs DRL with Human Feedback]

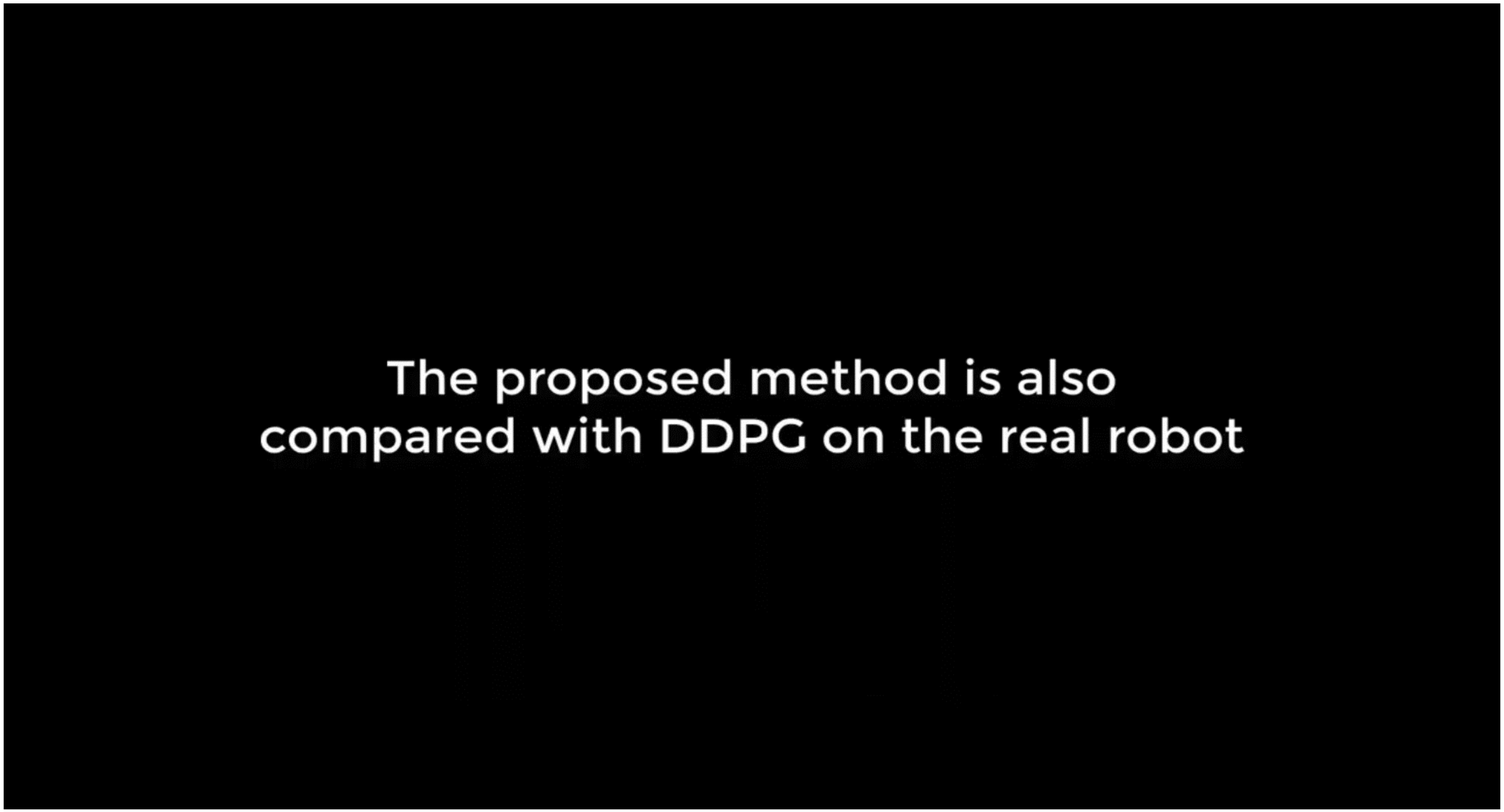


Source: https://arxiv.org/pdf/2102.10711.pdf;
https://github.com/hanlinniu/turtlebot3_ddpg_collision_avoidance

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Mobile Robot

Learning how to navigate while avoiding collisions by itself
[Deep Reinforcement Learning (DRL) vs DRL with Human Feedback]



Source: https://arxiv.org/pdf/2102.10711.pdf;
https://github.com/hanlinniu/turtlebot3_ddpg_collision_avoidance

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Humanoid Robots (2019)

Learning how to play soccer by themselves



Source: https://youtu.be/_PC-V5GJP6Q

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Humanoid Robots (2026)

Learning how to play soccer by themselves



https://www.youtube.com/shorts/0iSOeOa95PI

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Autonomous vehicle

Example 1: Wayve; Drive by self-learning

Alex Kendall, Co-Founder & CTO of Wayve says:

"**Our cars learn to drive from data with machine learning**. Every time a safety driver intervenes and takes over, the car learns to drive better. We don't tell the car how to drive, rather **it learns to drive from experience, example and feedback, just like a human**. This is **more safe and scalable than any other approach today**."

Source: https://techcrunch.com/2019/04/03/wayve-claims-world-first-in-driving-a-car-autonomously-with-only-its-ai-and-a-satnav/

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Autonomous vehicle

Example 1: Wayve; Drive by self-learning



Source: https://techcrunch.com/2019/04/03/wayve-claims-world-first-in-driving-a-car-autonomously-with-only-its-ai-and-a-satnav/

Challenges?

1. **Danger** to public during physical training
2. **Depreciation** of AV during physical training
3. **Still a L4; not L5!**

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Autonomous vehicle

Example 1: Wayve; Drive by self-learning together **with human feedback**



Source: https://www.youtube.com/watch?v=eRwTbRtnT1I

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Short Practice (30 mins)

Case Study:

Machinew wants their robotic arm system to pick and place target objects moving on a conveyor belt in an intelligent manner, i.e. pick from the conveyor belt and place into a bin. Given that RGB-Depth cameras (attached to the robotic arm) are used to identify the objects on the conveyor belt, you are tasked by Machinew to propose a suitable solution for their robotic arm system by considering **the following criteria:**

- The system's autonomous level is high (i.e. the robotic arm is able to make decision on their actions independently, with the help of the sensors and algorithms)
- The RGB-Depth cameras (attached to the robotic arm) are also used to detect the position and trajectory of the target objects, apart from just identifying them
- It is safe and able to avoid collision with other objects (e.g. non-target objects, other robots and/or any humans onsite). Assume that there are externally installed RGB-Depth cameras to detect the position and trajectory of these objects.
- The arm cannot be configured into odd positions that will risk damaging it (e.g. damaging the gripper when a bad grip point was chosen by the model, damaging the motor wires when certain set of joint angles are chosen).
- It can effectively and efficiently pick and place the target objects
- It utilizes multi-layer neural network(s) for its decision-making process

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Short Practice (30 mins)

<u>You are to:</u>

i. Identify 3 factors that are needed to be considered for your reward function based on the previously mentioned criteria,

ii. State how this could be undertaken by drafting an architecture to represent your solution. Note that your architecture must include the following components: the physical robot hardware, the target object, other objects to avoid collision with, the identification system, the decision-making system and other relevant systems.

iii. One issue that might arise during learning is using the real physical robot to train the model, which can be dangerous, considering that it would need real physical collisions to occur in order to properly train the robot to avoid them. Suggest with explanations what you can do to minimize the implications of this issue.

iv. Put your answers in PPT/WORD/PDF (your choice)

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